FW16 EXPANSION BAY –> MXM 3.x ADAPTER | schematic v4 DRAFT | generated + ERC–checked with KiCad 10 (kicad–cli)

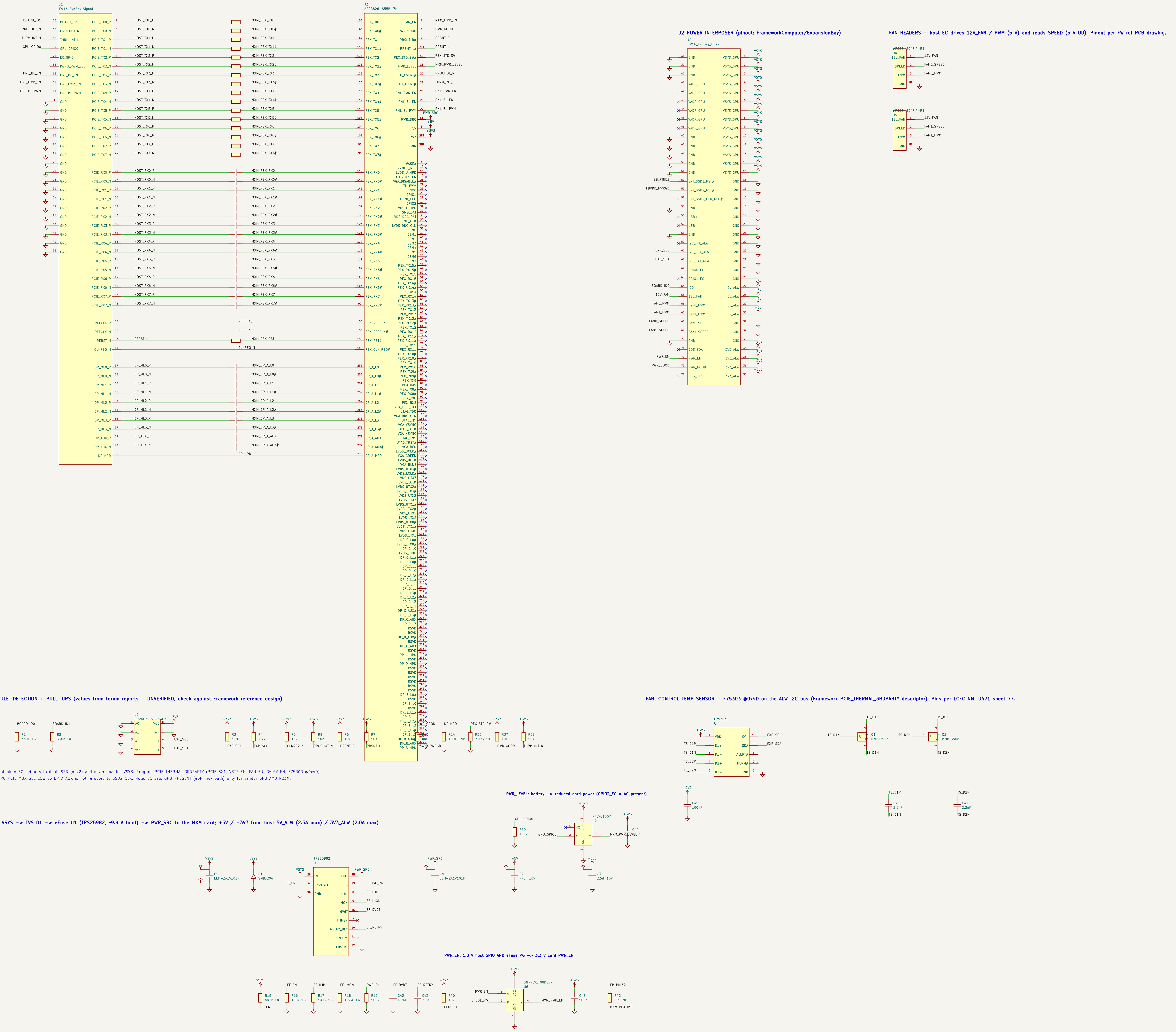
NOT A MANUFACTURABLE DESIGN. No PCB layout. Every value/pin/footprint marked TBD is an unresolved placeholder (see BOM CSV).

J1/J2 pins: github.com/FrameworkComputer/ExpansionBay Electrical/README.md ; J3: MXM 3.1 connector tables (ADLINK MXM–AXe, Advantech SKY–MXM–RTX3000).

Lanes go straight through by name: host PEX_TX –> MXM PEX_TX (MXM names are system–relative). Confirm against the MXM spec before ordering.

Series parts: host–>card PCIe R20–R35 = 0R; card–>host PCIe C10–C25 = 220nF; DP_A lanes+AUX C26–C35 = 100nF DC blocking; HPD direct (R14 pull–down DNP).

OPEN: TH_PWM unconnected. J3 = S55B on a 1.2 mm adapter; needs the flat 3D–printed bay cover (15.3 mm inside). Pins 52/53 and the F75303 address per FW16 7040 interface schematic. Checked against MXM 3.1 r1.0.



J2 POWER INTERPOSER (pinout: FrameworkComputer/ExpansionBay)

FAN HEADERS – host EC drives 12V_FAN / PWM (5 V) and reads SPEED (5 V OD). Pinout per FW ref PCB drawing.

EC MODULE–DETECTION + PULL–UPS (values from forum reports – UNVERIFIED, check against Framework reference design)

EEPROM: blank = EC defaults to dual–SSD (x4x2) and never enables VSYS. Program PCIE_THERMAL_3RDPARTY (PCIE_8X1, VSYS_EN, FAN_EN, 3V_5V_EN, F75303 @0x4D), but set GPU_PCIE_MUX_SEL LOW so DP_A AUX is not rerouted to SSD2 CLK. Note: EC sets GPU_PRESENT (eDP mux path) only for vendor GPU_AMD_R23M.

FAN–CONTROL TEMP SENSOR – F75303 @0x4D on the ALW I2C bus (Framework PCIE_THERMAL_3RDPARTY descriptor). Pins per LCFC NM–D471 sheet 77.

PWR_LEVEL: battery –> reduced card power (GPIO2_EC = AC present)

POWER: VSYS –> TVS D1 –> eFuse U1 (TPS25982, ~9.9 A limit) –> PWR_SRC to the MXM card; +5V / +3V3 from host 5V_ALW (2.5A max) / 3V3_ALW (2.0A max)

PWR_EN: 1.8 V host GPIO AND eFuse PG –> 3.3 V card PWR_EN

ERC + netlist checked with KiCad 10 kicad–cli. Not validated electrically. TBD = unresolved.

Sheet: /
File: fw16_mxm_adapter.kicad_sch

Title: Framework 16 Expansion Bay to MXM 3.x Adapter

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